

09 — Submodule + Tracking Hygiene (Ongoing)

Revision: 1 **Parent:** 00_MASTER_INDEX.md **Prerequisites:** None (can run anytime, recommended throughout all stages)

Overview

This stage covers submodule governance, tracking tooling, and documentation hygiene. These are NOT production-blocking individually but cumulatively represent significant technical debt. Perform throughout the project; do not batch at the end.

I-01 [AGENT] — Add helix-deps.yaml to All 6 ota-* Bricks

Effort: S (~30 min) **Source:** DELTA_ANALYSIS §2 SUB-NEW-1

Current state:

NONE of the 6 ota-* bricks (ota-protocol, ota-telemetry-schema, ota-artifact-validator, ota-rollout-engine, ota-update-engine-bridge, ota-android-agent) has a helix-deps.yaml. All other owned submodules DO have one.

What to do:

1. For each brick, create helix-deps.yaml:

```
# helix-deps.yaml – machine-readable dependency manifest
#(§11.4.31)
repo: HelixDevelopment/<brick-name>
type: library
```

```
language: go # or kotlin for ota-android-agent / ota-update-  
engine-bridge  
dependencies: []
```

2. Commit and push to each brick's origin.

I-02 [AGENT] — Add upstreams/ Recipes to All 6 ota-* Bricks

Effort: S (~30 min) **Source:** DELTA_ANALYSIS §2 SUB-NEW-1

Current state:

The 6 ota-* bricks have only a single origin remote. Per §11.4.36, each brick should push to all 4 mirrors (GitHub, GitLab, GitFlic, GitVerse).

What to do:

1. For each brick, create `upstreams/install_upstreams.sh` following the pattern from other submodules (e.g., `containers/upstreams/`).
 2. Run the script: `bash upstreams/install_upstreams.sh`.
 3. Verify: `git remote -v` shows all 4 remotes.
 4. Commit and push.
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I-03 [AGENT] — Add Wire Schema-Version Field

Effort: M (~2h) **Source:** DELTA_ANALYSIS §2 SUB-NEW-2

Current state:

Wire payloads carry no `SchemaVersion/ProtocolVersion` field. The only versioning is the coarse REST URL path `/api/v1`. Cross-version device ↔ server compatibility is unnegotiable.

What to do:

1. **ota-protocol:** Add `SchemaVersion int` to `UpdateAvailable`, `UpdateCheckRequest`, `TelemetryEvent`, `DeviceRegisterRequest`. Default to 1 for v1.0.0.
 2. **Server:** Accept `SchemaVersion: 1`, reject unknown versions with `400 UNSUPPORTED_SCHEMA_VERSION`.
 3. **ota-telemetry-schema:** Add `SchemaVersion int` to `TelemetryEvent`.
 4. **Kotlin DTOs:** Add `schemaVersion` field to DTOs in `ota-android-agent/core/.../Dtos.kt`.
 5. **Tests:** `TestSchemaVersionNegotiation`, `TestRejectsUnknownSchemaVersion`.
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I-04 [AGENT] — Resolve Mirror-Fork Canonicalization

Effort: M (~2h, after A-04 decision) **Source:** `PRODUCTION_READINESS_PLAN.md` K1

Affected submodules:

- `vision_engine` — 6 fixes at 1ec0c57, publish blocked
- `llm_orchestrator` — 3 fixes + 3 tests, scratchpad only
- `llm_provider` — 1 fix, scratchpad only

After A-04 resolves the canonical org + branch:

1. **Clone the canonical mirror** as the base.
 2. **UNION-merge** the divergent branches (never keep-ours, never force-push §11.4.113):
 - Apply the preserved fixes on top of the canonical base
 - Run tests on the merged tree
 - Push to all 4 mirrors from the canonical branch
 3. **Retire** non-canonical branches (tag as `archive/<branch>` if needed for history).
 4. **Bump parent gitlinks** in the main `helix_ota` repo to the new canonical commits.
 5. **Verify:** All 4 mirrors show the same HEAD for the canonical branch.
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I-05 [AGENT] — Adopt Constitution workable-items Engine

Effort: M (~2h) **Source:** DELTA_ANALYSIS §3 KI-2

Current state:

The in-repo `cmd/workable-items/main.go` is a simplified reimplementation. Its `db-to-md` sync is an explicit stub (`main.go:937-938`). The canonical engine lives at `constitution/scripts/workable-items/cmd/workable-items/`.

What to do:

1. **Remove** the local `cmd/workable-items/` directory (retire the simplified copy).
 2. **Reference** the constitution engine by path: `constitution/scripts/workable-items/cmd/workable-items`.
 3. **Update scripts** that call the local tool to call the constitution engine instead.
 4. **Verify:** `db-to-md` sync works bidirectionally (DB ↔ Issues.md ↔ Fixed.md byte-identical round-trip).
 5. **Per §11.4.177:** The constitution tooling is project-agnostic; it operates on the invocation directory — no hardcoded project path.
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I-06 [AGENT] — Extend `workable_items.db` Schema

Effort: M (~1h) **Source:** DELTA_ANALYSIS §3 KI-3

Missing columns/tables:

- `items` table missing: `created_by`, `assigned_to` (§11.4.104), `canonical_track + branch` (§11.4.181/§11.4.191), `reopens_count` (§11.4.55)
- Missing: `logic_groups` table (§11.4.191), `test_diary` table (§11.4.149)

What to do:

1. **Add columns:**

```

ALTER TABLE items ADD COLUMN created_by TEXT NOT NULL DEFAULT
    '';
ALTER TABLE items ADD COLUMN assigned_to TEXT NOT NULL DEFAULT
    '';
ALTER TABLE items ADD COLUMN canonical_track TEXT NOT NULL
    DEFAULT '';
ALTER TABLE items ADD COLUMN branch TEXT NOT NULL DEFAULT '';
ALTER TABLE items ADD COLUMN reopens_count INTEGER NOT NULL
    DEFAULT 0;

```

2. Add tables:

```

CREATE TABLE IF NOT EXISTS logic_groups (
    group_name TEXT PRIMARY KEY,
    track TEXT NOT NULL,
    branch TEXT NOT NULL UNIQUE,
    description TEXT NOT NULL DEFAULT ''
);

CREATE TABLE IF NOT EXISTS test_diary (
    id TEXT PRIMARY KEY,
    item_id TEXT NOT NULL REFERENCES items(ota_id),
    run_date TEXT NOT NULL,
    verdict TEXT NOT NULL CHECK (verdict IN
        ('PASS', 'FAIL', 'SKIP')),
    evidence_path TEXT NOT NULL DEFAULT '',
    notes TEXT NOT NULL DEFAULT ''
);

```

3. Run migration via scripts/testing/migrate_workable_items.sh.

I-07 [AGENT] — Fix §11.4.33 Type-Aware Closure Vocabulary

Effort: S (~30 min) **Source:** DELTA_ANALYSIS §3 KI-4

Current state:

12 closed items use the wrong closure vocabulary: - 7 Features marked as Fixed → should be Implemented (→ Fixed.md) - 5 Tasks marked as Fixed → should be Completed (→ Fixed.md) - Only 1 Bug (OTA-020) uses the correct Fixed (→ Fixed.md)

What to do:

```
UPDATE items SET status = 'Implemented (→ Fixed.md)' WHERE type =  
    'Feature' AND status = 'Fixed (→ Fixed.md)';  
UPDATE items SET status = 'Completed (→ Fixed.md)' WHERE type =  
    'Task' AND status = 'Fixed (→ Fixed.md)';
```

Verify:

```
sqlite3 docs/workable_items.db  
    "SELECT type, status, count(*) FROM items WHERE status  
    LIKE '%Fixed.md%' GROUP BY type, status"  
  
# Should show:  
#   Bug|Fixed (→ Fixed.md)|1  
#   Feature|Implemented (→ Fixed.md)|7  
#   Task|Completed (→ Fixed.md)|5
```

I-08 [AGENT] — Reconcile Track/Branch Drift + Record Canonical Bindings

Effort: S (~30 min) **Source:** PRODUCTION_READINESS_PLAN.md §2.9

Current drift:

- config/multitrack/the-factory.yaml: T2 = web dashboard/website, T3 = containers-hardening, T4 = vm-emu-hardening
- Operator directive: claude1 → T2 accounts, claude3 → T4 website
- Accounts delivery plan: canonical branch feature/multi-tenant-accounts (not feature/accounts-web)

What to do:

1. **Reconcile:** Update the-factory.yaml to match the canonical bindings:

- T2 / feature/multi-tenant-accounts (Accounts)
- T3 / feature/containers-hardening (Containers)
- T4 / feature/website (Website)

2. **Record in registry:** Insert into logic_groups table (from I-06):

```
INSERT INTO logic_groups VALUES ('multi-tenant-accounts',  
    'T2', 'feature/multi-tenant-accounts', 'Multi-tenant  
Accounts feature');  
INSERT INTO logic_groups VALUES ('containers-hardening', 'T3',  
    'feature/containers-hardening', 'Containers  
hardening');  
INSERT INTO logic_groups VALUES ('website', 'T4', 'feature/  
website', 'Marketing website');
```

3. **Enforce:** The §11.4.181 gate (CM-BRANCH-NAME-CONSISTENCY) fails if ANY submodule touched by a logic group uses a divergent branch name.
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I-09 [AGENT] — Bump llm_orchestrator Parent Gitlink

Effort: S (~10 min, after I-04) **Source:**
PRODUCTION_READINESS_PLAN.md §2.6

Current state:

Parent gitlink 7e3e6da lags the submodule HEAD by 2 commits. After I-04 resolves the mirror fork and the canonical HEAD is established, bump the gitlink:

```
cd submodules/llm_orchestrator  
git pull origin main # or canonical branch  
CANONICAL_HEAD=$(git rev-parse HEAD)  
cd ../../  
git add submodules/llm_orchestrator  
git commit -m "chore: bump llm_orchestrator gitlink to  
    $CANONICAL_HEAD"
```

I-10 [AGENT] — Fix Stale Comments

Effort: S (~20 min) **Source:** PRODUCTION_READINESS_PLAN.md K5, K6

K5 (VAL-1):

File: submodules/ota-artifact-validator/stages.go:15 **Current:** Comment claims ota-protocol.ValidateArtifactMeta enforces a 255-char Version bound. **Fix:** Change comment to name the real in-package ValidateVersion guard.

K6 (fabric-lease):

Files: server/internal/store/postgres_fabric.go:137-139, store.go:220-229 **Current:** Comments still describe old non-exclusive lease behavior (fixed by HB-2 03340896). **Fix:** Update comments to describe current uniform exclusive-lease behavior.

I-11 [AGENT] — Remove LLMPProvider Symlink

Effort: S (~10 min) **Source:** DELTA_ANALYSIS §3 KI-5

Current state:

submodules/LLMPProvider is a symbolic link to llm_provider (NOT a duplicate directory as OTA-027 originally framed).

What to do:

1. **Verify nothing resolves submodules/LLMPProvider path:**

```
grep -r "LLMPProvider\|LLMPProvider" --include="*.go" --  
include="*.mod" --include="*.yaml" server/  
submodules/ 2>/dev/null
```

2. **If no references:** Remove the symlink:

```
rm submodules/LLMPProvider
```

3. **If references exist:** Update them to use llm_provider (the tracked submodule), then remove.
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Verification Checklist

Step	Action	Expected Result
I-01	helix-deps.yaml in all 6 bricks	Each brick has the manifest
I-02	upstreams/ in all 6 bricks	4 remotes per brick
I-03	Schema version field	Wire payloads carry version, server rejects unknown
I-04	Mirror forks resolved	All submodules canonical, fixes published
I-05	Constitution engine adopted	db-to-md sync works bidirectionally
I-06	DB schema extended	New columns + tables exist
I-07	Closure vocabulary fixed	12 items retyped correctly
I-08	Track/branch bindings recorded	logic_groups table populated
I-09	llm_orchestrator gitlink bumped	Parent points to canonical HEAD
I-10	Stale comments fixed	Comments match current behavior
I-11	LLMProvider symlink removed	No stray references

Honest Boundary (§11.4.6)

- Steps I-01 through I-03 are independent of any other stage and can be done immediately.
- Step I-04 (mirror forks) is gated on operator decision A-04. The fixes exist (committed or scratchpad-preserved) but cannot be published until canonical mirrors are chosen.

- Steps I-05 and I-06 (workable-items tooling) are tooling improvements — they don't block production but should be done before the next release.
- Step I-07 (closure vocabulary) is a data fix — no code changes needed.
- Step I-11 (symlink) was verified this session: `submodules/LLMProvider` is indeed a symlink (not a duplicate directory).